Bike Station Analysis Reports & Insights

- **Keerthana S**

The dataset represents the operational data of a Smart City Bike Sharing system used across several European cities. It contains information about bike stations, their geographic locations, parking capacity, bike availability, and supported payment methods. This data helps analyze how efficiently the bike-sharing infrastructure supports urban transportation.This data helps analyze bike station utilization, infrastructure capacity, and accessibility of bike-sharing services across cities**.**

Dataset Attributes

Column	Description
Station Name	Name of the bike sharing station
City	City where the station is located
Bike Stand Capacity	Total number of parking slots available at the station
Available Bikes	Number of bikes currently available for rental
Available Bike Stands	Number of empty slots available for returning bikes
Payment Type	Indicates whether the station supports cash or digital payment methods
Bonus Available	Is There Any bonus available for particular station in cities

Data Cleaning

- **Standardized text formatting** to maintain consistency across station names, cities, and categorical fields.
- **Removed null, duplicate, and incomplete records** to improve data quality and reliability.

Before Cleaning

A	B	C	D	E	F	G	H	I	J	K	L	M	N
number	name	address	position	banking	bonus	status	Contract N	Bike Stand	Available E	Available E	Last Update		
2000	2000 - TEST DSI PLAIS		46.548876	False	False	OPEN	maribor	1	0	0	2022-06-02T15:14:38+05:30		
1297	1297- ST F	ST FERREC	43.293485	True	False	OPEN	marseille	9	0	0	2022-04-06T12:54:35+05:30		
8265	8265- NÉG	NEGRESKC	43.269318	True	False	OPEN	marseille	15	0	0	2022-09-27T10:57:48+05:30		
8065	8065-PROI	PROMENA	43.267272	True	False	OPEN	marseille	15	15	0	2022-06-08T11:50:13+05:30		
1011	01011 - BC	Borne test	47.19504,	False	False	OPEN	nantes	0	0	0	2022-05-23T20:11:50+05:30		
4342	4342 - FLA	FLAMMAR	43.305993	True	False	OPEN	marseille	10	0	0	2022-09-13T11:28:01+05:30		
6176	6176-CAN	CANTINI R	43.284775	True	False	OPEN	marseille	24	24	0	2022-09-06T18:02:17+05:30		
3322	3322 - ARE	32 rue de (	43.312337	True	False	OPEN	marseille	20	13	7	2022-09-14T13:27:09+05:30		
1264	1264 - CAN	CANEBIER	43.297896	True	False	OPEN	marseille	10	0	0	2022-09-12T11:18:28+05:30		
1191	1191-CAN	CANEBIER	43.296207	True	False	OPEN	marseille	20	2	0	2022-10-03T11:34:01+05:30		
5058	5058 -SAK	SAKAKINI	43.292800	True	False	OPEN	marseille	10	10	0	2022-09-13T15:41:24+05:30		
4293	4293- BLA	BLANCARD	43.296263	True	False	OPEN	marseille	14	0	0	2022-09-26T16:44:57+05:30		
1245	1245-GAM	GAMBETT	43.299027	True	False	OPEN	marseille	15	0	0	2022-10-03T12:02:43+05:30		
7153	7153-COR	CORSE SAI	43.289467	True	False	OPEN	marseille	10	0	0	2022-10-10T12:40:37+05:30		
6154	6154-PIER	PIERRE PU	43.290329	True	False	OPEN	marseille	8	0	0	2022-09-26T11:29:34+05:30		
1002	1002-HOT	HOTEL DE	43.296138	True	False	OPEN	marseille	32	0	0	2022-10-17T11:36:42+05:30		
1224	1224-PLAI	PLAINE BI	43.295825	True	True	OPEN	marseille	16	0	0	2022-10-24T13:08:08+05:30		
1287	1287- STAI	STALINGR	43.299520	True	False	OPEN	marseille	17	0	0	2022-10-04T11:49:03+05:30		
500	500 - IT PLAISIR		48.799828	False	False	OPEN	jcdecauxbi	1	0	1	2022-08-30T21:00:02+05:30		
5220	5220-ROO	ROOSEVEL	43.298991	True	True	OPEN	marseille	10	0	0	2022-11-08T14:53:04+05:30		
6160	6160-153 I	153 PARAD	43.286838	True	False	OPEN	marseille	14	0	0	2022-11-22T12:47:11+05:30		

After Cleaning

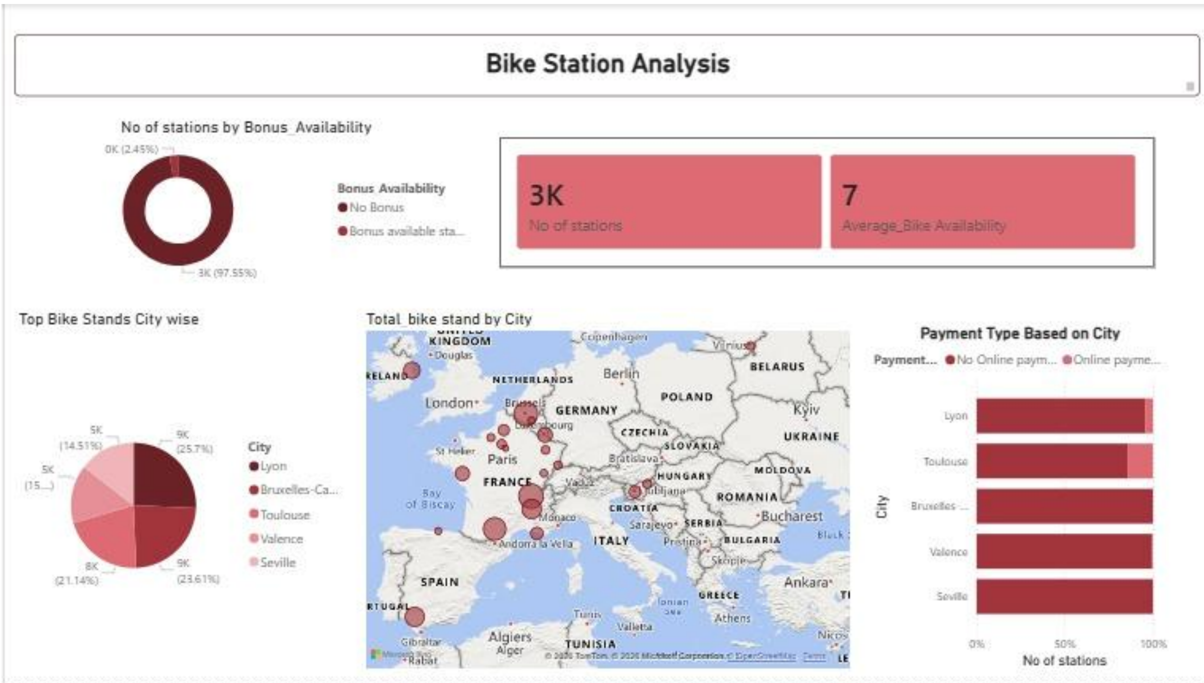
Bike Station Name	banking	bonus	status	City	Bike Stands	Available Bike Stands	Available Bikes
Novembre	FALSE	FALSE	OPEN	Lyon	35	1	34
Gaston Berger	FALSE	FALSE	OPEN	Lyon	35	0	35
Novembre Négogousses	FALSE	FALSE	OPEN	Toulouse	18	6	10
Liberation	TRUE	TRUE	CLOSED	Marseille	10	10	0
Paradis	TRUE	FALSE	OPEN	Marseille	14	0	0
Maisons Charles V (Cb)	TRUE	FALSE	OPEN	Nancy	23	23	0
Mazargues	TRUE	FALSE	OPEN	Marseille	15	0	0
Michelet	TRUE	FALSE	CLOSED	Marseille	19	17	0
Prado	TRUE	FALSE	CLOSED	Marseille	13	10	0
Racine	FALSE	FALSE	OPEN	Lyon	30	28	0
Août	FALSE	FALSE	OPEN	Lyon	15	12	1
Août / Pressensé	FALSE	FALSE	OPEN	Lyon	19	10	8
Atelier Vélo'V	TRUE	FALSE	OPEN	Lyon	3	1	1
Septembre (Cb)	TRUE	FALSE	OPEN	Besancon	15	9	6
Corniche Kennedy	TRUE	FALSE	CLOSED	Marseille	15	15	0
Libération	TRUE	TRUE	CLOSED	Marseille	10	10	0
De-Octubre-Cieza	FALSE	FALSE	OPEN	Valence	20	16	4
A Rilspert	FALSE	FALSE	OPEN	Luxembourg	15	7	6
A Schommesch	FALSE	FALSE	OPEN	Luxembourg	20	14	4
A. Vienuolio	FALSE	FALSE	CLOSED	Vilnius	18	0	0
Abbaye De La Cambre / Abdij Ter Kameren	FALSE	FALSE	OPEN	Bruxelles-Capitale	20	15	5
Adeps	FALSE	FALSE	OPEN	Bruxelles-Capitale	22	11	7
Adolfo Suarez	FALSE	FALSE	OPEN	Valence	30	2	26
Adolphe Max	FALSE	FALSE	OPEN	Bruxelles-Capitale	21	9	11
Aeropolis	FALSE	FALSE	OPEN	Bruxelles-Capitale	25	7	18
Agora	FALSE	FALSE	OPEN	Bruxelles-Capitale	19	6	12
Al Bréck	FALSE	FALSE	OPEN	Luxembourg	15	8	4
Alameda De Hercules	FALSE	FALSE	OPEN	Seville	20	10	10
Alameda De Hercules	FALSE	FALSE	OPEN	Seville	20	5	14
Alameda De Hércules	FALSE	FALSE	OPEN	Seville	20	11	9

Dax Measures

```
Average_Bike_Availability = AVERAGE(BikeTable[Available Bikes])
No of stations = COUNTROWS(BikeTable)
```

Dax Custom Column

```
PaymentType = IF(BikeTable[banking] = True,"Online Payment Available","Cash Only”
Bonus Availability = IF(BikeTable[bonus]=True,"Yes Discount Available","No Discount")
```



From the dashboard, the bike-sharing system consists of around 3,000 bike stations distributed across several European cities.
On average, each station **has about 7 bikes available**, which indicates that the network maintains a moderate supply of bikes at any given time.

From a city mobility perspective, this suggests the system is designed for **high circulation rather than high storage**. Bikes are expected to move frequently between stations rather than remain idle.

Key Insight

- The system prioritizes **continuous usage and redistribution** instead of keeping large bike inventories at each station.
- Maintaining an average of **7 available bikes** indicates operational efficiency but also signals potential **demand pressure during peak hours**.

The map visualization shows that bike stations are concentrated across major European urban centers including:

- Lyon
- Toulouse
- Brussels
- Valencia
- Seville

Among these, Lyon appears to have the highest station concentration, indicating that the city has a stronger adoption of the bike-sharing infrastructure.

Interpretation

Cities with larger station networks typically:

- Experience higher demand for micro-mobility
- Have better cycling infrastructure
- Promote eco-friendly transportation policies

City-wise Bike Stand Capacity

Key Observation

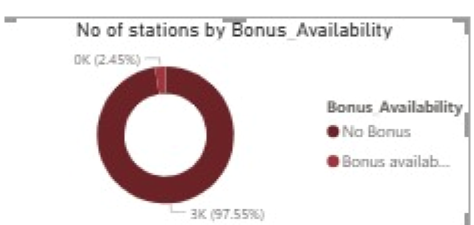
Lyon and Brussels together account for nearly half of the bike stand capacity.

This indicates that these two cities are the primary operational hubs of the bike-sharing program.

Operational Meaning

- Higher rider demand
- Denser station networks
- Larger urban populations using shared mobility

Bonus Availability at Stations



The bonus availability chart shows that nearly all stations provide bonus incentives, while only a very small number do not.

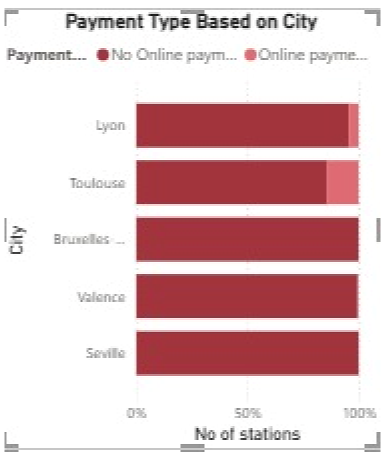
- Bonus systems in bike sharing typically encourage riders to:
- Return bikes to less crowded stations
- Help with natural bike redistribution
- Reduce operational costs for manual redistribution

Insight

The heavy use of bonus incentives suggests that the system relies on user behavior to balance bike availability across stations**.**

This is a smart operational strategy because it reduces the need for costly logistics operations.

Payment Method Adoption



While comparing with Offline payments online payments Method are less available so improving digitalized payment will help to improve the surface.

Conclusion

The bike-sharing system demonstrates a well-developed smart mobility infrastructure across multiple European cities. With over **3,000 stations**, strong digital payment adoption, and incentive-based redistribution mechanisms, the system is designed to support high-frequency urban commuting.

However, the concentration of infrastructure in a few key cities suggests opportunities to expand capacity and optimize bike distribution, ensuring consistent availability as demand grows.